Geophys. J. R. astr. Soc. (1980) **60**, 263–281

Lower Cretaceous magnetic stratigraphy in Umbrian pelagic carbonate rocks

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Received 1979 June 22; in original form 1979 May 17

Summary. A record of geomagnetic field polarity for the Barremian, Aptian and Albian stages of the Early Cretaceous has been derived in three overlapping sections of pelagic carbonate rocks in the Umbrian Apennines of northern Italy. The remanence carrier in the greyish-white Majolica limestone and Fucoid Marls is magnetite, with haematite also an important constituent in a zone of 'couches rouges' within the Fucoid Marls. The weak remanent magnetizations were measured with a cryogenic magnetometer. Alternating field or thermal demagnetization was used to isolate the characteristic remanent magnetization (ChRM) in 655 specimens from 248 stratigraphic levels. The samples respond positively to a tectonic fold test, indicating that the ChRM predates the Late Tertiary folding of the Umbrian sequence. The magnetic stratigraphy derived from variations of virtual geomagnetic pole latitude clearly defines the recognizable reversal pattern associated with Mesozoic marine magnetic anomalies M0 to M4. The sections have been zones palaeontologically on the basis of planktonic foraminifera and calcareous nannofossil assemblages. The ages of magnetic anomalies M0 to M4 determined in this way are somewhat older than those in the reversal time scale of Larson & Hilde (1975). Anomaly M0 is located in the Early Aptian, close to the Aptian/Barremian boundary. A long period of normal polarity in the Aptian and Albian corresponds to the early part of the Cretaceous magnetic quiet zone. It is interrupted in the Late Aptian by a reversal which we find in only one of the Fucoid Marl sections, and which has not been reported in oceanic magnetic anomaly investigations.

Institut für Geophysik, ETH Zürich, Contribution Nr. 242

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1 Introduction

The record of geomagnetic polarity during the Mesozoic and Cenozoic is known best from interpretation of the magnetic anomaly lineations recorded in oceanic crust formed during this time, since the break-up of Pangaea in the Late Triassic. The largest part of this polarity record has not yet been confirmed by independent palaeomagnetic investigations. The youngest portion of the magnetic time scale of Heirtzler *et al.* (1968) correlates excellently with the polarity sequence determined in radiometrically dated lavas younger than about 3.5 Myr (Doell, Dalrymple & Cox 1966). The older Cenozoic and Late Cretaceous anomalies were dated by simple linear extrapolation from this short baseline. Although McDougall *et al.* (1977) have extended recently the baseline length to about 6.5 Myr in stratigraphically sequential lavas, the geomagnetic polarity record for the remainder of the Cenozoic has not yet been verified. For most of this period the errors inherent in radiometric dating prohibit development of a magnetic reversal sequence from stratigraphically unrelated samples. On the other hand recent major improvements in magnetometer technology now permit rapid and accurate investigation of sedimentary sequences whose remanent magnetizations were previously measurable only with great difficulty or not at all. Particularly well suited for magnetic stratigraphy studies are sequences of pelagic carbonate rocks.

The sequence of geomagnetic reversals derived from the Late Cretaceous anomalies 34 to 29 has been confirmed recently in the Scaglia Rossa pelagic limestone at Gubbio in the Northern Apennines of Italy (Lowrie & Alvarez 1977a, b) and verified in a nearby check section at Moria (Fig. 1) (Alvarez & Lowrie 1978). The reversals were correlated to the Campanian and Maastrichtian stages by palaeontological zonation of the Gubbio section (Premoli Silva 1977) which has been established as a magnetostratigraphic type section for the Late Cretaceous (Alvarez *et al.* 1977). The Gubbio and Moria sections also confirmed the existence of a long period of normal geomagnetic polarity preceding anomaly 34 which could be correlated with the Cretaceous magnetic quiet zone.

According to the oceanic anomaly record the onset of the Cretaceous quiet interval marks the termination of the Mesozoic reversal sequence (Larson & Pitman 1972). The youngest anomaly of this sequence is designated M0 and has been assigned a Late Aptian age (Larson & Hilde 1975). Recent magnetic stratigraphy studies in the Southern (Vicentinian) Alps by Channell, Lowrie & Medizza (1979) have cast doubt on this date. They derived a magnetic stratigraphy in the Early Cretaceous Biancone limestone, in which M0 was dated at the base of the Aptian.

The present investigation was undertaken to confirm and date the youngest reversals of the Mesozoic sequence, in particular to determine the age of anomaly M0. These new studies have been carried out in three partially overlapping sections of pelagic carbonate rocks at Poggio le Guaine, Valdorbia and Gorgo a Cerbara in the Umbrian Apennines of northern Italy (Fig. 1). Data from a single site in the Valdorbia section were incorporated in a composite polar wander path for Umbria by VandenBerg, Klootwijk & Wonders (1978). This Late Aptian site was reversely magnetized, but we will show that this Valdorbia reversal does not correspond to anomaly M0.

2 Lithostratigraphy and sampling scheme

In the Umbrian sequence the Late Cretaceous is represented by the greyish-white, Cenomanian to mid-Turonian, Scaglia Bianca limestone and the pink-to-red Scaglia Rossa limestone which extends from the mid-Turonian through the remaining Late Cretaceous stages and, indeed, was deposited until the Middle Eocene. The magnetic stratigraphy of these pelagic limestones was developed and checked in the Gubbio and Moria sections (Lowrie & Alvarez 1977a, b; Alvarez & Lowrie 1978).

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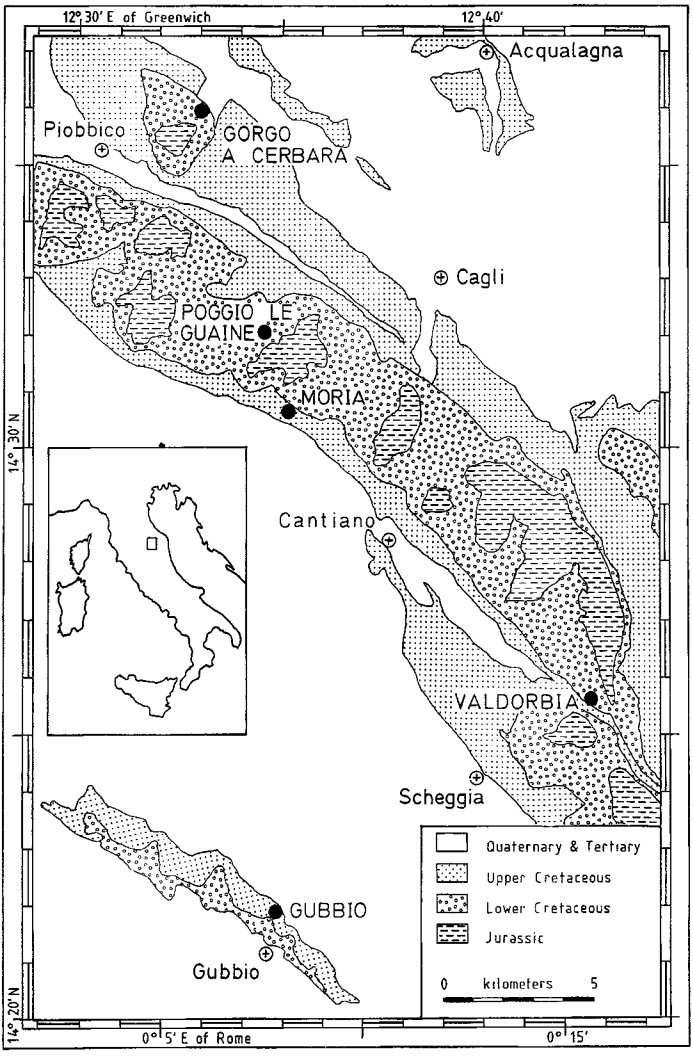


Figure 1. Location map of the northern Umbrian Apennines. The Lower Cretaceous pelagic carbonate sediments for the present study were sampled from the sections at Gorgo a Cerbara, Poggio le Guiane and Valdorbia. The magnetic stratigraphy of the Upper Cretaceous was determined in earlier studies of limestone sections at Gubbio and Moria.

Late Aptian and Albian stages are represented in the Umbrian sequence by marls, with bedding planes occasionally marked by fucoids ('Scisti a Fucoidi'). These Fucoid Marls consist of an upper grey member and a lower red member ('couches rouges'), both of which are characterized by indurated beds from a few centimetres to about 15 cm in thickness, separated by marly intervals that can be up to a few metres thick, and which can only rarely be sampled. The Fucoid Marls were sampled in overlapping sections at all three locations (Fig. 2).

The Early Cretaceous stages up to and including the Early Aptian are represented in the Umbrian sequence by the Majolica limestone, which is the lithologically-similar, Umbrian equivalent of the Biancone limestone that was successfully investigated by Channell *et al.* (1979) in the Southern Alps. It is a compact, white, pelagic limestone containing black chert

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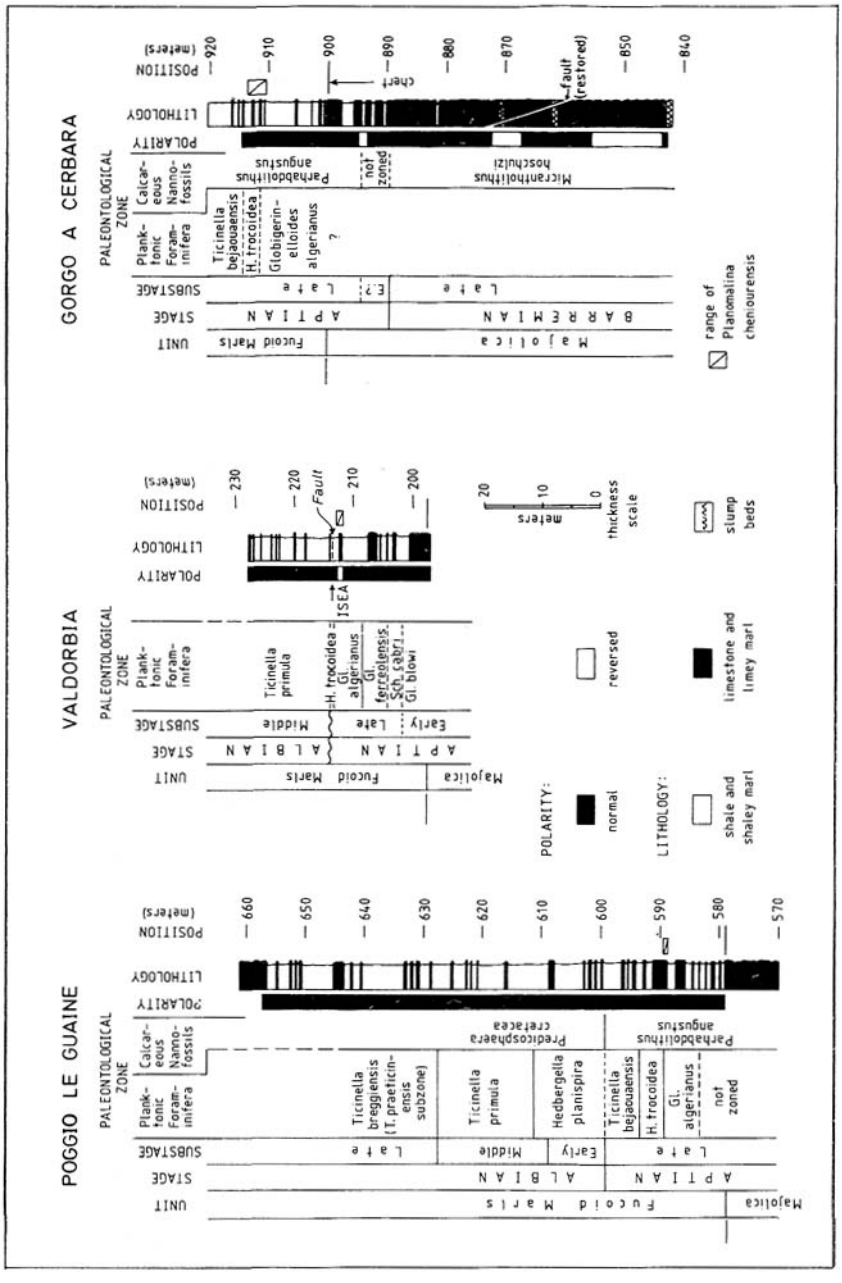


Figure 2. Stratigraphy and biostratigraphy of the Poggio le Guaine, Valdorbia and Gorgo a Cerbara sections. The metre-positions are referred to arbitrary zero in each section. Abbreviated forms for planktonic foraminifera are: Gl. = Globigerinelloides, H. = Hedbergella, T. = Ticinella, Sch. = Schackoina.

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beds, whose name derives from its porcelain-like appearance on fresh-broken surfaces (the Majolica lacks the large planktonic foraminifera which cause the Scaglia to break on surfaces speckled with little bumps). The Majolica limestone was sampled only in the Gorgo a Cerbara section; frequent soft-sediment slumping and faulting disturb the Majolica in Umbria and only at this location could we be sure of good control of these tectonic complications.

Preliminary sampling of the three sections was based on a combination of oriented hand samples and drilled cores. Later, more intensive sampling was carried out by taking closely spaced drilled cores in zones of particular interest. Only the indurated beds were sampled, and as a result the sample distribution is somewhat irregular in the Fucoid Marls, although where possible the sampling interval was kept to less than a metre.

2.1 THE POGGIO LE GUAINE SECTION

The Poggio le Guaine section is located on the flat-dipping, crestal part of an anticlinal mountain near the town of Cagli (Fig. 1). Although the Scisti a Fucoidi and Scaglia are both severely disturbed a short distance (0.5 km) to the northeast, the section itself is apparently undisturbed. The Fucoid Marl beds are flat-lying and well-exposed, and could be sampled up to the gradational transition into the overlying Scaglia Bianca; in this section faulting is minor and compensatable. The Scaglia Bianca/Fucoid Marl contact is placed at the lowest occurrence of chert nodules in the Scaglia. The Fucoid Marls, which are free of chert, continue downward through the middle and lower Albian into the upper Aptian. Sampling of the section was terminated at the Fucoid Marl/Majolica contact. This contact is placed at the highest occurrence of chert nodules in the Majolica, and falls in the upper Aptian. Fifty-two beds were sampled, yielding 180 25-mm cylindrical palaeomagnetic specimens. The underlying Majolica limestone at this location forms a precipitous cliff which was not sampled because of the difficulty of access.

2.2 THE VALDORBIA SECTION

The Valdorbia section lies in a roadcut beside the Scheggia–Sassoferato road, 5 km north-east of Scheggia (Fig. 1). The sampled interval covers only the lower part of the Fucoid Marls, which are severely disturbed in their upper part; a fault of unknown throw is found in the middle of the section. However, although it lies close to a syncline and thrust, the section is continuous and undisturbed upward from the Majolica for 30 m into the Fucoid Marls, and is definitely not overturned. The ISEA reversed site of VandenBerg *et al.* (1978) is located in this section. The beds dip fairly uniformly at about 55° to the south-west. A total of 24 oriented hand samples yielded 97 palaeomagnetic specimens. The underlying Majolica limestone is also well-exposed here, but is covered by steel netting to prevent loose blocks from falling on to the roadway.

2.3 THE GORGO A CERBARA SECTION

The Gorgo a Cerbara section is located in the bed of the Candigliano River, 4 km east of the town of Piobbico (Fig. 1). The upper grey member of the Fucoid Marls is intensely distorted but the lower 'couches rouges' member and the Majolica lie in conformable contact with one another and are excellently exposed, with a uniform dip of about 50° to the north-east. The section is in fact on the flank of an anticline whose Early Jurassic core is exposed along the river. Although the Majolica is well exposed in the stream bed for a thickness of several hundred metres, synsedimentary slumping and subsequent faulting are common.

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From 860 to 874 m the section is cut by a fault. However, the offset could be determined and allowed for by matching two distinctive slump beds, each about 1 m thick, across the fault. Although the presence of slump beds indicated some penecontemporaneous disturbance of the section, this appears to be minor, and it does not interrupt the continuity of the nannoplankton zonation. At 842.5 m, we encountered a disturbed zone parallel to bedding; since we could not be sure of continuity of section below this level, we terminated sampling at this point. A total of 187 hand and drilled samples yielded 378 specimens from this section.

2.4 SEDIMENTATION RATES AND SAMPLING INTERVAL

The sedimentation rates in the different sections were calculated by determining the thickness of sediment deposited in each stage (whose boundary positions were determined palaeontologically as described in the next section) and assigning to each stage boundary the age given by Sigal (1977). The Albian and Upper Aptian Fucoid Marls at Poggio le Guaine are about 70 m thick, and span a time interval of some 12 Myr. Their mean sedimentation rate is around 6 m Myr^{-1}. The Lower Aptian and Barremian parts of the Majolica limestone at Gorgo a Cerbara are about 60 m thick and represent some 10 Myr. Here again the mean sedimentation rate is about 6 m Myr^{-1}. It should be noted that much higher sedimentation rates (around 25 m Myr^{-1}) are necessary lower in the section to account for the tremendous thickness of Majolica limestone extending to the Cretaceous–Jurassic boundary.

The average sedimentation rate of 6 m Myr^{-1} allows us to estimate approximately the time intervals represented by our sampling. Within each standard one-inch specimen data are averaged for up to 4000 yr. In some parts of the sections the sampling interval was a few centimetres, representing about 10 000 yr; in other stratigraphic intervals, particularly in the marly grey member of the Fucoid Marls at Poggio le Guaine, the sampleable indurated beds were occasionally as far as 3 m apart, a gap equivalent to about 500 000 yr. If reversals of the geomagnetic field occurred during deposition of the few large marly gaps they may have been missed in our sampling.

3. Palaeontological zonation

In the period of time corresponding to deposition of the Fucoid Marls (mainly Aptian–Albian) planktonic foraminifera can be abundant and rich in diagnostic forms on whose distribution a detailed zonation has been established (Sigal 1977; Premoli Silva, Paggi & Monechi 1976). Earlier than the Aptian, planktonic foraminiferal assemblages are very rare, poorly diversified and of little use for a detailed biostratigraphy. This is particularly true for facies such as the Majolica. Calcareous nannoplankton are the best represented and the most useful fossils in the Lower Cretaceous. Their biostratigraphic resolution is not very high (biozones are quite comprehensive), however some appearances and extinctions within the calcareous nannofossil assemblages correlate well with stage boundaries (Thierstein 1973).

In the present paper foraminifera were studied by Premoli Silva both on washed material and in thin section, while calcareous nannoplankton were studied by Monechi with optical and scanning electron microscopes. The planktonic foraminiferal and calcareous nannoplankton biozenes identified in the three sections are plotted and correlated with stages in Fig. 2.

Planktonic foraminifera occur discontinuously throughout the Fucoid Marls due to heavy dissolution in the middle and basal parts of the formation; some levels are completely carbonate-free. Consequently some biozones could not be identified. Dissolution affected the nannoplankton also; however, the phenomenon did not prevent identification of the zones.

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3.1 POGGIO LE GUAINE SECTION

In the Poggio le Guaine Section, the Scaglia Bianca/Fucoid Marls contact is placed at the lowest occurrence of chert nodules. The highest palaeontological sample came from 7 m below this level and yielded a planktonic foraminiferal fauna belonging to the *Ticinella breggiensis* Zone (*T. praeticinensis* Sub-zone) correlated with the lower part of the Upper Albian. The remaining part of the Fucoid Marls is attributed to the Middle and Lower Albian and Upper Aptian. The Albian/Aptian boundary is located on the basis of the appearance of the nannolith *Predicosphaera cretacea,* whereas planktonic foraminifera are poorly represented.

The contact between Fucoid Marls and Majolica is placed at the highest occurrence of chert nodules. On the basis again of the nannoplankton content, it falls in the Upper Aptian. Planktonic foraminifera present in the basal part of the Fucoid Marls are poorly diagnostic.

3.2 VALDORBIA SECTION

The Valdorbia Section covers only the lower part of the Fucoid Marls. Planktonic foraminiferal zones from the Lower and Upper Aptian and Middle Albian were recognized. Nannofossils are not diagnostic in this section. A possible fault within the shaly interval at about 214 m removes the latest Aptian, the Early Albian and part of the middle Albian. The faunal change is very abrupt and leaves no doubt about the ages above and below the discontinuity.

The reversed polarity zone is located in indurated beds just below, but clear of, the fault. We use the site designation ISEA of VandenBerg *et al.* (1978) who first reported this Late Aptian reversed direction, which is associated with a fossil of narrow range, *Planomalina cheniourensis.*

3.3 GORGO A CERBARA SECTION

The Gorgo a Cerbara Section includes the lower part of the Fucoid Marls. Three planktonic foraminiferal zones are identified in the highest levels sampled at this locality; all are attributed to the Late Aptian. On the basis of the nannoplankton content the lowermost part of the Fucoid Marls and the upper levels of the underlying Majolica are also Late Aptian in age. The Late Aptian *Parhabdolithus angustus* Zone (Thierstein, 1973) was identified down to 891.95 m. Sampling in the Gorgo a Cerbara Section continued for about 60 m downwards into the Majolica. In this interval the age was established on the basis of the nannofossil content alone. The lowermost part of the *P. angustus* zone contains the second reversed polarity zone which is here correlated with anomaly M0 (see below). Below the *P. angustus* zonal boundary a short interval (from 891.95 to 889.15 m) could not be assigned to any standard zone (all diagnostic forms are missing); it could represent the Lower Aptian. We saw no evidence for a hiatus. However, if a hiatus exists in this part of the section, this interval could be the lowermost part of the Upper Aptian.

From 889.15 m down to the base of the usable section at 842.65 m the nannoplankton assemblage is characterized by *Nannoconus colomi* and other nannoconids. This assemblage is indicative of the Barremian *Micrantholithus hoschulzi* Zone. The Aptian–Barremian boundary is thus located at about 890 m.

3.4 SUMMARY

The following palaeontological conclusions are of importance in dating the reversal sequence recognized in the Fucoid Marls and the upper part of the Majolica (and described in the following sections). The Fucoid Marls cover the interval from the Upper Albian down to the

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Upper Aptian. This unit is entirely of normal polarity, with the exception of an ambiguous reversed zone which was found only in the Valdoria section, where it falls near the top of the Late Aptian. The Fucoid Marl/Majolica contact falls in the Upper Aptian. The M0 reversed zone probably corresponds to the Lower Aptian, and the other reversed zones encountered (M1 and M3) fall in the Barremian.

4 Remanent magnetizations

The remanent magnetizations of the 655 specimens were measured with a three-axis ScT cryogenic magnetometer. Each magnetization was measured twice and the individual components of the replicate measurements were combined as described by Heller (1977) so as to give a synthetic dispersion parameter (ψ_{63}). In Fig. 3 can be seen how this dispersion parameter increases as the remanent intensity decreases and the instrumental noise increases.

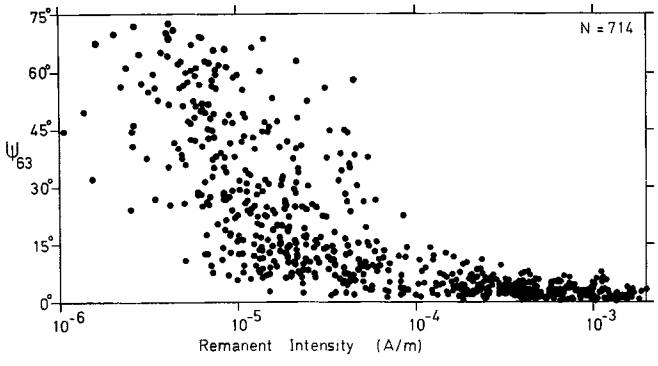


Figure 3. Variation of the precision parameter ψ_{63} with intensity of remanent magnetization in 714 measurements on Lower Cretaceous pelagic carbonate rocks. ψ_{63} is similar to the angular standard deviation of repeated measurements and is high when the intensity or quality of the data is low. The parameter is a useful empirical guide to the reliability of the progressively weakening magnetization at successive stages of magnetic cleaning.

Large values also result from operator mistakes and the presence of unstable remanence components whose values alter between the replicate measurements. The value of ψ_{63} serves as a guide to the reliability of each magnetization measurement and can be used as a quantitative, if somewhat arbitrary, criterion for rejecting poorly determined data. In the present study all magnetizations weaker than 5×10^{-6} A m^{-1} or having a ψ_{63} value greater than about 30° were systematically eliminated from further consideration.

The intensity of the natural remanent magnetization (NRM) is dependent upon the lithology (Fig. 4). It is greatest in the 'couches rouges' member of the Fucoid Marls, and drops by about a factor of 50 in the Majolica limestone. Within each of the two lithologies there is no systematic fluctuation of NRM intensity like that observed in the Scaglia Rossa and Scaglia Bianca by Alvarez & Lowrie (1978).

Each specimen was demagnetized progressively in peak alternating fields of 5, 10, 15 and 20 mT. Additionally, a small number of specimens was demagnetized in higher fields, but beyond about 40 or 50 mT the residual remanences in Majolica specimens became too weak to measure satisfactorily. The least stable directions and weakest remanences were found in the Majolica limestone, where a very large low-coercivity magnetization component was present, and the directional stability was poorer than in the Fucoid Marls (Fig. 5). The grey

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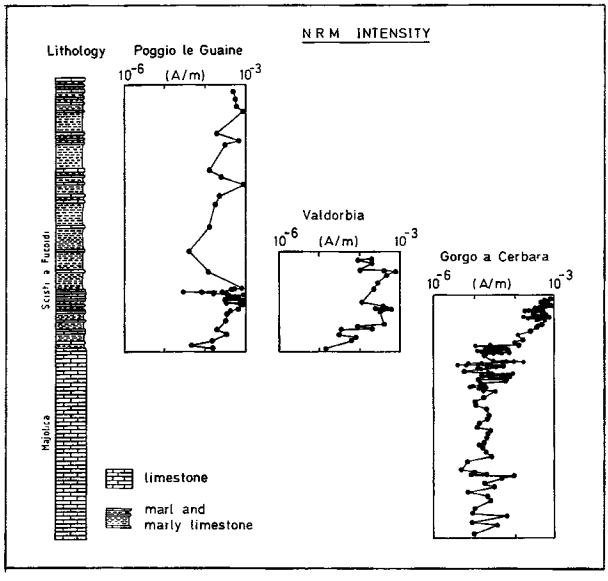


Figure 4. The remanent intensity of the pelagic carbonate rocks was closely dependent on their lithology. It was highest in the Fucoid Marls and decreased by a factor of 50 or more to extremely low intensities in the Majolica limestone in which most of the Early Cretaceous reversals were found.

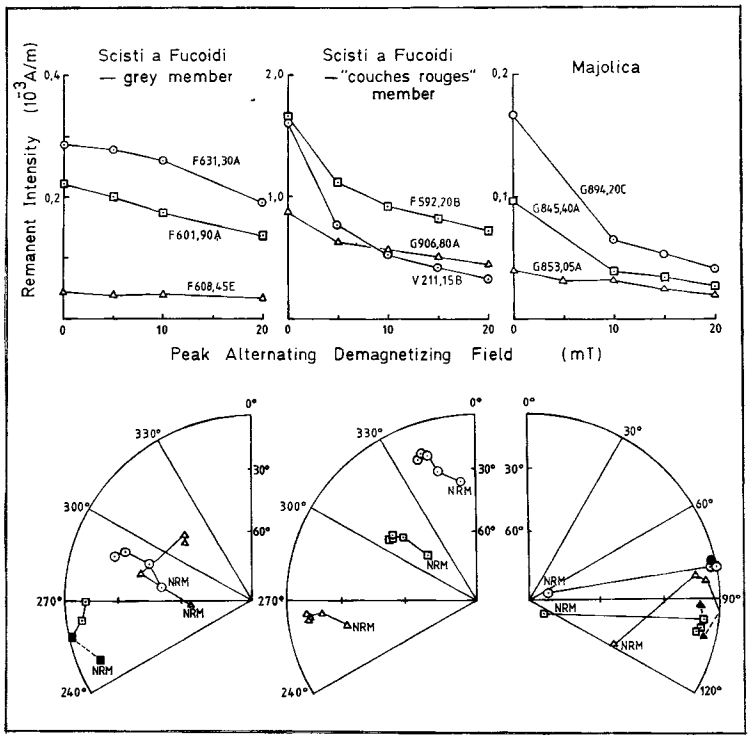


Figure 5. Alternating field demagnetization behaviour of representative samples of the grey and 'couches rouges' members of the Fucoid Marls (Scisti a Fucoidi) and Majolica limestone, showing intensity and directional stability characteristics.

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member of the Fucoid Marls also had a prominent 'soft' component, but after its removal the direction was fairly stable. The red 'couches rouges' samples were more strongly magnetized and, after removal of an initial large soft component, whose effects were most pronounced in the negatively magnetized samples, the remanence was stable in intensity and direction.

Selected specimens were given isothermal remanent magnetization (IRM) in progressively stronger fields. The total coercivity spectrum was deduced by differentiating the IRM acquisition curves. In specimens of the grey member of the Fucoid Marls (Fig. 6) and in Majolica specimens saturation IRM was reached in fields of less than 0.4 T. Clearly the magnetic mineralogy in these pelagic carbonate rocks is dominated by the presence of a magnetic mineral which is relatively easily saturated and which, by analogy to studies of the Scaglia limestones (Lowrie & Alvarez, 1977b), we interpret to be magnetite. In such limestones there is little reason to suspect that important secondary magnetizations might be present. This is not the case for the 'couches rouges' samples. IRM curves (Fig. 6) have a strong component below 0.3 T but are not saturated in fields of 1 T; in addition to magnetite in these beds we infer that haematite is present, which causes their red colour and may carry an important secondary magnetization component.

The rate of change of direction was computed at each stage of the demagnetization curve for each specimen using a method similar to that of Symons & Stupavsky (1974) and the 'most stable' direction was selected as the characteristic remanent magnetization (ChRM) for the specimen. The ChRM directions of specimens from the same sample were compared for consistency, then combined vectorially to get the mean direction for the sample. Because of excessive directional inconsistency all specimens from 15 samples were rejected from further palaeomagnetic analysis.

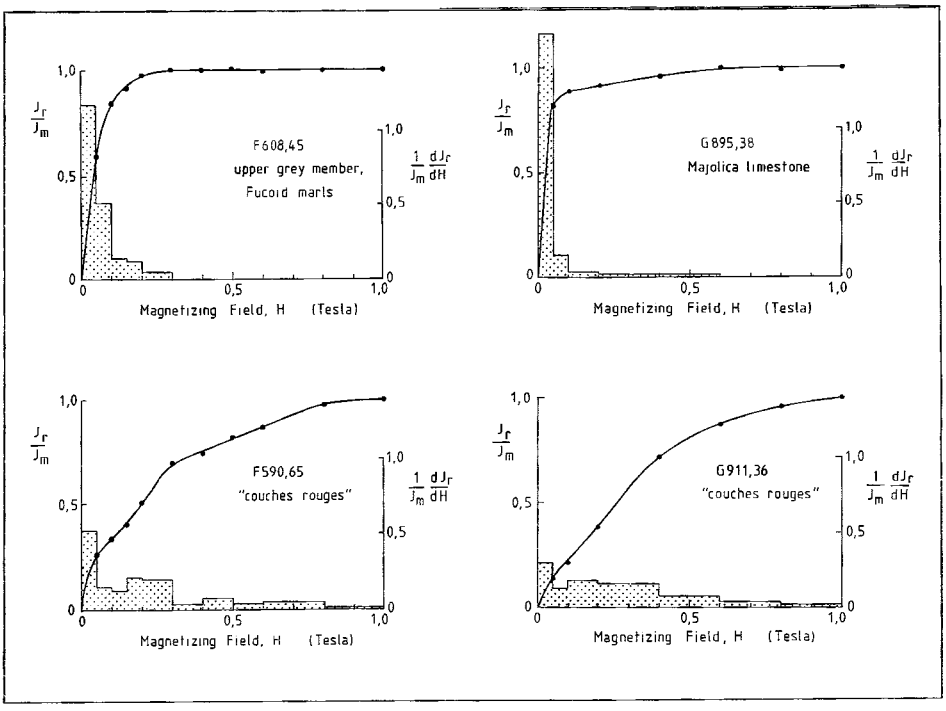


Figure 6. Acquisition curves of isothermal remanence (J_r), normalized in terms of the maximum value (J_m) reached in the experiment. The shaded histograms show the coercivity spectra obtained by differentiating the acquisition curves with respect to the applied field.

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Samples with exclusively normal ChRM directions were found at the Poggio le Guaine site, both normal and reversed ChRM directions occurred at the other two locations. The normal equivalent directions of reversely magnetized samples were used to construct equal area density stereograms of the ChRM directions before and after AF cleaning, and after making appropriate tectonic corrections at each location.

The NRM directions (Fig. 7a) are grouped noticeably better than the directions after AF cleaning. This is due to the presence of a low coercivity component whose direction is close to that of the present field in Umbria and which probably represents a recent viscous remanence. This soft component is removed by the AF cleaning, so that the ChRM directions at first appear to be dispersed (Fig. 7b). They fall into three distinct groups, each corresponding to a different sampling location; the group of northerly directions come from the Valdorbia

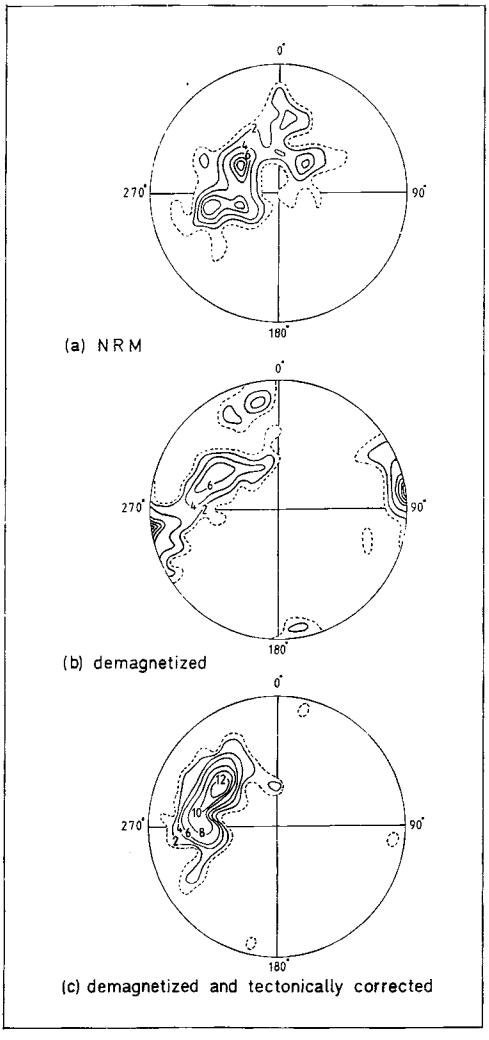


Figure 7. Lower hemisphere, equal area density maps of (a) the natural remanent magnetization (NRM) directions, (b) the optimally demagnetized, characteristic remanent magnetization (ChRM) directions and (c) the ChRM directions after local tectonic corrections. Reversed polarities have been inverted and plotted on a common hemisphere with normal directions. The numbers on the contour lines give the point density as a percentage per unit area.

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section, the westerly directions are found at Gorgo a Cerbara, and the north-westerly directions come from Poggio le Guaine. There is not enough variation of bedding attitude within each section to affect the within-site dispersion of ChRM directions. However, after correcting for the SW dip of the beds at Valdorbia and the NE dip of the beds at Gorgo a Cerbara, the directions from these two locations move together into good agreement with the directions from the flat-lying beds at Poggio le Guaine (Fig. 7c). Except for the Valdorbia Fucoid Marls the inclinations of the mean directions of each section are very similar after tectonic correction but there are small differences in the declinations of the mean directions (Table 1), which become more noticeable when the data are grouped according to biostratigraphic age.

For example, identical average inclinations of 44° are found in the Late Aptian in all three sections, but the average declinations range from 300 to 325° (Table 2). The Early Albian mean directions at Poggio le Guaine and Valdorbia also differ mainly in declination.

Table 1. *Mean directions of characteristic remanent magnetization after tectonic correction for each formation at each sampling locality.* α_{95} *and* k *are the radius of the circle of confidence (in degrees) and the precision parameter, respectively.*

FORMATION	SAMPLING LOCALITY	THICKNESS SAMPLED (m.)	NUMBER OF SAMPLES	MEAN DIRECTION DECL.	INCL.	STATISTICAL PARAMETERS α_{95}	k
Fucoid Marls:							
grey member	Poggio le Guaine	58	15	301	38	5	58.7
	Valdorbia	12	9	310	29	13	16.2
couches rouges	Poggio le Guaine	22	37	311	44	5	20.8
	Valdorbia	15	15	327	40	17	6.04
	Gorgo a Cerbara	15	72	306	42	5	11.1
Majolica limestone	Gorgo a Cerbara	58	100	282	44	5	8.37

Table 2. Average palaeomagnetic directions in the three Umbrian sections, grouped according to biostratigraphic age determinations. N = number of sampled stratigraphic levels; D, I = mean declination and inclination, respectively; α_{95} and k as in Table 1.

	POGGIO LE GUAINE					VALDORBIA					GORGO A CERBARA				
	N	D	I	α_{95}	k	N	D	I	α_{95}	k	N	D	I	α_{95}	k
ALBIAN															
Late	9	301	41	5.7	82	-	-	-	-	-	-	-	-	-	-
Early	6	301	33	9.7	48	10	312	30	12	17	-	-	-	-	-
APTIAN															
Late	37	311	44	5.3	21	13	325	44	19	5.6	114	300	44	4.6	9.4
Early	-	-	-	-	-	-	-	-	-	-	13	283	48	8.6	24
BARREMIAN	-	-	-	-	-	-	-	-	-	-	45	280	42	8.7	6.9

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The declinations in the Albian and Late Aptian Fucoid Marls are distinctly larger than the westward declinations in the Early Aptian and Barremian Majolica limestone (Tables 1 and 2).

After rotating the pole of the stereogram to the mean direction of Fig. 7c, the distribution is found to be approximately elliptical with an axial ratio of about 1.5 : 1. As a result the tectonically-corrected ChRM directions in Fig. 7c appear more smeared in declination than in inclination. At least three factors may contribute to this elongation of the combined directional distribution. Firstly, intermediate directions at all three locations, especially at reversal boundaries, have not been removed. Secondly, the mean ages of the samples are slightly different at each section. Poggio le Guaine and Valdorbia samples are Albian and Late Aptian in age and Gorgo a Cerbara samples are Aptian and Barremian. The difference between the mean directions of the sections (Table 1) may result from polar wander from the Barremian through the Albian; little is known about geomagnetic field behaviour during these Early Cretaceous stages, but the observed trend is compatible with expectations based on either the African or European polar wander path of Van der Voo & French (1974).

The third factor which could contribute to the smearing of directions in Fig. 7c is incorrectness of the tectonic corrections. The beds at each site were restored to the horizontal by assuming that they had been folded simply about a horizontal axis. Although the folding of the Umbrian Apennines appears at first to be simple, the tectonic style may actually be more complex than hitherto suspected (Alvarez, Engelder & Geiser 1978). Considerable divergence of palaeomagnetic directions throughout Umbria may be related to local variation in fold axis attitude (Channell *et al.* 1978) and may account for distinct differences in declination between long sections that are less than 20 km apart (Alvarez & Lowrie 1978).

The positive response to the tectonic fold test indicates that ChRM directions in the Fucoid Marls and Majolica limestone predate the Late Tertiary folding of the Umbrian Apennines. The dominance of magnetite in the grey member of the Fucoid Marls and in the Majolica argue in favour of a depositional (or immediately post-depositional) origin for the remanent magnetization, as found in the Scaglia Rossa and Scaglia Bianca pelagic limestones elsewhere in Umbria (Lowrie & Alvarez 1975, 1977b). It is therefore permissible to associate the remanent magnetizations with their stratigraphic positions, although the effect of hematite in the 'couches rouges' samples must be considered in more detail.

5 Magnetic stratigraphy

The declinations and inclinations of tectonically corrected ChRM directions showed a systematic relationship to stratigraphic position (Fig. 8). All Poggio le Guaine samples were normally magnetized, and there was no appreciable difference of directions between grey member and 'couches rouges' member samples. A clearly indicated reversal in the 'couches rouges' member at Valdorbia is marked by simultaneous inversion of both declinations and inclinations. The reversed directions are not exactly antipodal to the normal samples; the angular difference between the mean directions of the two groups is 130°. It is probable that a stable secondary magnetization has not been removed by the AF cleaning. This reversal corresponds to the reversely magnetized ISEA site of VandenBerg *et al.* (1978). No corresponding reversed interval is found in the 'couches rouges' samples from the other sections, but several synchronous inversions of declination and inclination are present in the Majolica in the Gorgo a Cerbara section.

The latitude of the virtual geomagnetic pole (VGP) relative to the sampling sites at the time of deposition is used as a clearer indicator of geomagnetic polarity. The inclination of

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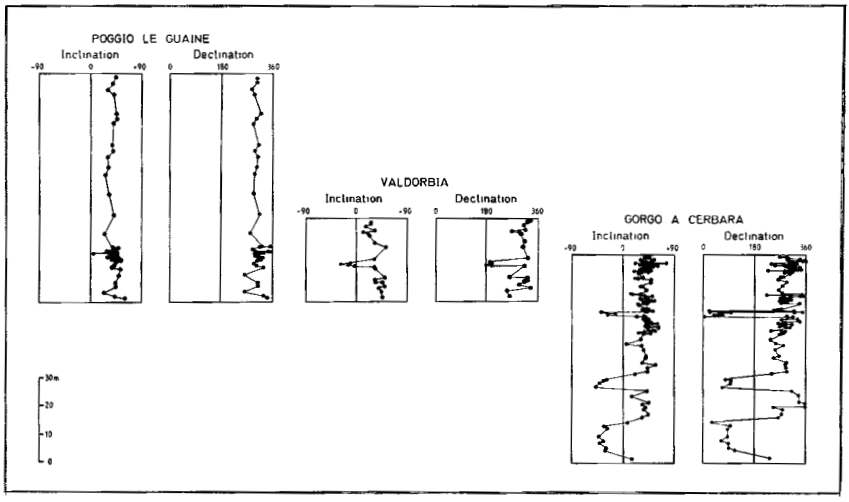


Figure 8. Stratigraphic variations of declination and inclination of the characteristic remanent magnetization (ChRM) in each section.

the site mean direction serves to compute the latitude of the site during deposition, the declination relative to the site mean declination is substituted for the actual declination and is used with the observed inclination to compute the instantaneous relative VGP latitude for each sample in each section. The variation of VGP latitude defined an unambiguous magnetic stratigraphy characterized by sharply delineated zones of normal and reversed polarity from which the history of geomagnetic polarity can be inferred for this part of the Early Cretaceous (Fig. 9). This polarity sequence was correlated with the biostratigraphic zonations

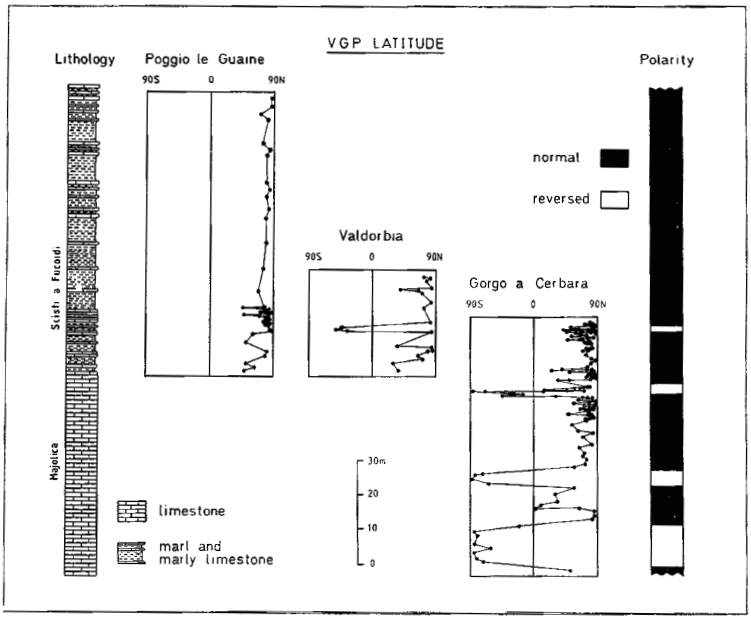


Figure 9. The virtual geomagnetic pole (VGP) latitude has been derived for each sample by equating the pole position derived from the mean direction of each section with the Earth's rotation axis. The VGP latitude fluctuations of individual samples relative to the mean value are interpreted in terms of geomagnetic polarity in the right-hand column.

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of the three sections (Fig. 2) so as to date each of the polarity reversals. No reversals are found in the Albian, two occur in the Aptian, and several are found in the Barremian. Further interpretation of these reversals requires a clearer understanding of the nature of the ISEA reversed polarity zone.

6 The Valdorbia Reversal (ISEA)

The directions of magnetization in red 'couches rouges' samples at Poggio le Guaine and Gorgo a Cerbara were normal, whereas those in four adjacent 'couches rouges' beds at Valdorbia were reversed. The absence of reversals in the first two sections may imply that the indurated beds were not originally deposited synchronously. The IRM acquisition tests suggest the presence of both magnetite and haematite in 'couches rouges' samples. Therefore, after demagnetizing them to 25 mT, the Valdorbia reversely magnetized samples were further demagnetized by stepwise progressive thermal demagnetization.

Demagnetization data obtained from pelagic limestones are usually of sufficiently good quality to permit analysis of directional variation. Consistency of directions or directional trends can usually be established before signals become too weak or unstable to measure accurately. In samples from their reversed site ISEA VandenBerg *et al.* (1978) established two distinct components of remanence. A component between 0.11 and 0.3 T was identified as the primary component; the lower coercivity component could not be identified. In our samples from the Valdorbia section the following characteristics are clearly established (Fig. 10): (1) a large directional change occurs during AF demagnetization to 10 mT, accompanying removal of a normal overprint; (2) the directions at 15 and 25 mT are stable and their mean

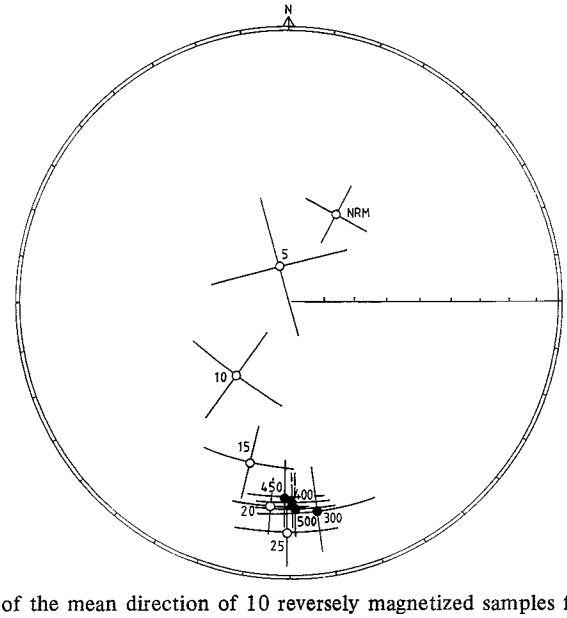


Figure 10. Variation of the mean direction of 10 reversely magnetized samples from the reversal in the Valdorbia section, prior to tectonic corrections. All directions are on the lower hemisphere; numbers 5–25 give the peak AF demagnetizing field in millitesla; numbers 300–500 identify the temperature (°C) to which the samples were heated in subsequent thermal demagnetization. The bars on each point define the orthogonal diameters of the 95 per cent confidence circle. The directions after thermal demagnetization in the 300–500°C range are indistinguishable from those after 20 and 25 mT AF demagnetization; all have negative polarity after tectonic correction.

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has declination 188°, inclination −14°; (3) during further thermal demagnetization the directions remain quite stable and over the range 300–500°C the mean direction has declination 184°, inclination −12°; (4) accurate measurements were not possible after heating to 600°C, because the remanences became at the same time weak and unstable, and were randomly directed.

We interpret from these results that the magnetite and haematite components in the Valdorbia samples have essentially the same stable directions, and that this corresponds to reversed polarity.

It is unlikely that the haematite formed long after deposition. The next confirmed geomagnetic reversal after deposition of the Valdorbia sediment corresponds to the negative Gubbio A-zone (Alvarez *et al.* 1977) which occurred more than 30 Myr later, by which time the sediment was buried to a depth of more than 300 m. In order to be negatively magnetized like the magnetite fraction the 'couches rouges' haematization probably occurred fairly soon after deposition as a result of the decomposition of goethite during diagenesis. It is not greatly erroneous to associate the reversal in the Valdorbia 'couches rouges' beds with the stratigraphic level at which it is found.

7 Discussions

Correlation of magnetic stratigraphies from one section to another, or correlation of the magnetic stratigraphy derived in a sedimentary section with the geomagnetic polarity sequence deduced from oceanic anomalies, is only possible when the record possesses a distinctive character, or 'fingerprint'. Otherwise, the cross-correlation of similar binary sequences presents unresolvable ambiguity. The appearance of the 'fingerprint' can be drastically altered by non-uniformity of sedimentation rate and/or sea-floor spreading rate. The magnetic reversal history of the Late Cretaceous between anomalies 34 and 29 contained alternating polarity zones of variable duration, and correlation of the magnetic stratigraphies in the Gubbio and Moria sections (Alvarez & Lowrie, 1978) and to the oceanic anomaly sequence (Lowrie & Alvarez, 1977a,b) was unmistakable. The oceanic equivalent to the Lower Cretaceous magnetic stratigraphy defined in the present study is to be found among the youngest anomalies of the Mesozoic sequence (Larson & Hilde 1975). The geomagnetic polarity pattern for anomalies M0 to M4 offers a suitable 'fingerprint' for correlation with the polarity sequence in Fig. 9.

The oldest reversal found in the Gubbio and Moria sections corresponded to anomaly 34, and occurred very near to the Santonian–Campanian boundary. The Santonian, Coniacian, Turonian and Cenomanian stages all belong to the Cretaceous magnetic quiet interval. From the present study there is no evidence of reversals in the Albian, and we infer that this stage also belongs to the quiet interval.

If the Valdorbia reversal ISEA is assumed to be a synsedimentary record of a Late Aptian geomagnetic reversal, the magnetic stratigraphy in the Lower Cretaceous pelagic carbonate rocks contains evidence for at least one extra reversal that has not been identified in the Mesozoic sequence. Two possible correlations of our data with the M0 to M4 anomaly sequence may then be suggested (Fig. 11). In correlation A the Valdorbia reversal is equated with the reversed zone that produces anomaly M0, and by implication the unidentified reversal lies between M0 and M1. It is, however, unlikely that such a prominent reversal would be overlooked in interpreting the oceanic anomalies. The correlation B has less distorted tie-line and identifies anomaly M0 with the Early Aptian; according to this correlation the Valdorbia reversal is an unidentified Late Aptian event. If the Valdorbia reversal is in fact a

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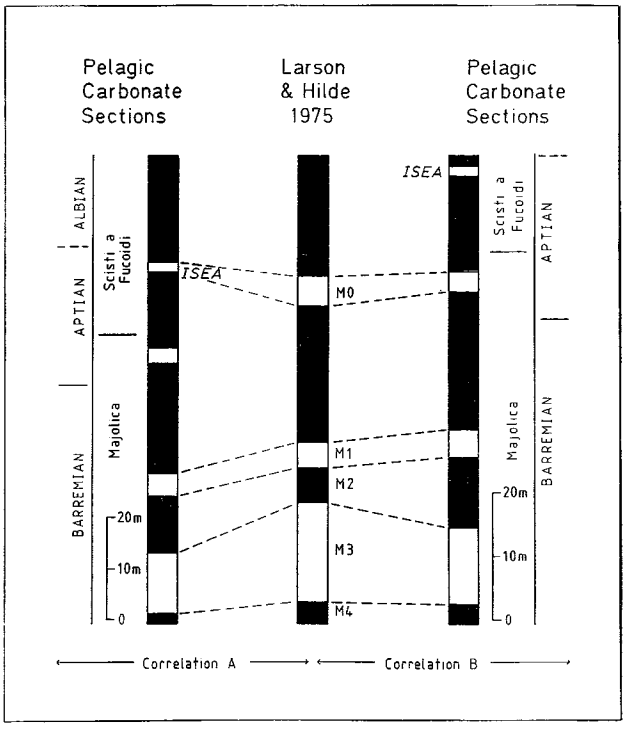


Figure 11. Without heed to other factors there are two most likely correlations of the polarity sequence in the pelagic carbonate sections with the polarity sequence derived from Mesozoic oceanic magnetic anomalies by Larson & Hilde (1975). In correlation A the Valdorbia reversal (ISEA) is equated with magnetic anomaly M0; in the favoured correlation B it is younger than M0.

much later remagnetization of the 'couches rouges' beds in the Valdorbia section, it would have to be rejected from the geomagnetic polarity sequence derived from the magnetic stratigraphy of Fig. 9, and in this case correlation B would again ensue.

8 Conclusions

The magnetic stratigraphies from the Upper Cretaceous Scaglia limestones at Gubbio and Moria are combined with those from the Lower Cretaceous Fucoid Marl and Majolica limestone sections in the present study to give a composite geomagnetic polarity sequence for the Barremian through Maastrichtian stages. This sequence is in good agreement with the geomagnetic polarity history of the Cretaceous as derived from oceanic magnetic anomalies (Fig. 12). According to this correlation anomaly M0 is of Early Aptian age, which agrees with the identification and dating of this anomaly in the Biancone limestone by Channell *et al.* (1979). These data imply that the Cretaceous magnetic quiet zone began earlier, ended later and consequently was of longer duration than has been assumed from interpretation of oceanic magnetic anomalies.

The identification of the Valdorbia reversal as a real geomagnetic phenomenon contemporaneous with deposition of the 'couches rouges' beds rests largely on rock magnetic grounds and is not entirely satisfactory. The reversal was not found in the 'couches rouges' member in the other sections, despite intensive re-sampling of all available beds. It is possible that the indurated beds are not time-equivalent in the different sections. Nevertheless, the reversal is

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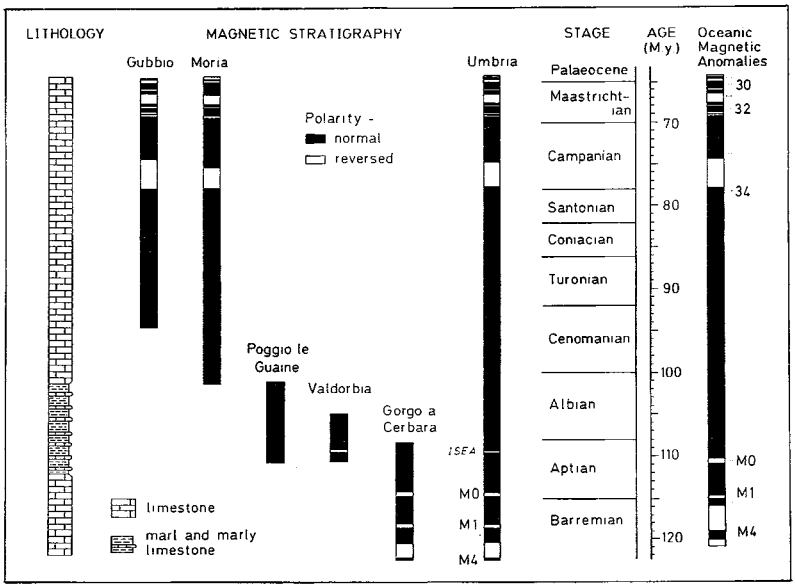


Figure 12. The magnetic stratigraphies of five Umbrian pelagic carbonate sections have been combined to give a composite polarity sequence for Umbria covering the Late, Middle and part of the Early Cretaceous. The Late Cretaceous oceanic magnetic anomaly sequence (Larson & Pitman 1972) has been correlated directly with the Gubbio section. The Early Cretaceous oceanic magnetic anomaly sequence has been plotted according to the numeric ages allocated to these reversals (Larson & Hilde 1975). Numeric ages associated with biostratigraphic stage boundaries are taken from the Cretaceous time scale of Van Hinte (1976).

found in three successive beds in the Valdorbia section, and the best argument in favour of its reality would be to detect it in another section. If it is real, this Late Aptian reversal ought to be recorded as a lineated magnetic anomaly within the presently defined Cretaceous magnetic quiet zone, just younger than anomaly M0.

Acknowledgments

We thank several colleagues for their help in this project, in particular Jim Channell, Friedrich Heller, Fabio Medizza and Lucia Paggi for their advice and criticism, and Marcia Lowrie for typing the manuscript. Partial financial support is gratefully acknowledged from US NSF grant EAR-76-03272 and Italian CNR Comitato 05 grants to Palaeopelagos Units, Florence and Milan.

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